

Inverse Spherical Dual Hemisphere Quantum Mechanics: Electromagnetic Operations and Absorption Fields

NOS v18 — Emission, Absorption, and the Bridge Factor

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Abstract

This document presents electromagnetic operations within the inverse spherical dual hemisphere framework established in NOS v17. Rooted in the foundational identity $\sqrt{1} = 1 = 512 \text{ bins} = 720^\circ$, electromagnetic phenomena emerge from the gearbox mechanics where $\pi_{\text{machine}} = 201/64 = 3.140625$ and $\alpha_{\text{raw}}^{-1} = 137$ arise from the same geometric position through ratio and difference operations. The bridge factor $\lambda_B = 137/128 \approx 1.0703$ connects matter field baselines to observed radiation values, with its inverse $\lambda_B^{-1} = 128/137 \approx 0.9343$ governing absorption. The four dual-hemisphere operators (dsin, dcos, csin, ccos) handle emission (decompression) and absorption (compression) processes across the mass-energy cycle. Emission operates at the Q1→Q2 transition (decompression dominant), while absorption operates at the Q3→Q4 transition (compression dominant). The seam at $d = 0^\circ$ provides the equilibrium pivot where both hemispheres meet. Energy unity is maintained across all quadrants: the same energy breathes through 1D (quantum), 2D (electromagnetic), 3D (thermal), and 4D (mass) manifestations without summation. Practical examples including hydrogen ground state, Lyman- α transitions, and circuit power demonstrate how observed values are recovered from matter field baselines through bridge factor conversion.

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Part I

The Foundational Identity and Gearbox Origin of Electromagnetic Constants

The Starting Point — The Closed Whole

All electromagnetic mechanics begin with the logical identity:

$$\sqrt{1} = 1 = 512 \text{ bins} = 720^\circ$$

This is the complete, unpartitioned whole. Electromagnetic emission and absorption are partition operations on this single closed sphere through the four dual-hemisphere operators.

The Gearbox — Origin of Electromagnetic Constants

At the critical 90° wall (Q1→Q2 transition), the partition structure creates two orthogonal readings from the same geometric position:

Flat radial lattice count: exactly 64 bins.

Curved arc count demanded by inverse closure: exactly 201 bins.

Two operations on the same position:

$$\pi_{\text{machine}} = \frac{201}{64} = 3.140625 \quad (\text{ratio operation})$$

$$\alpha_{\text{raw}}^{-1} = 201 - 64 = 137 \quad (\text{difference operation})$$

The fine-structure constant $\alpha = 1/137$ is not an empirical mystery — it is the geometric difference between curved and flat partition counts at the 90° wall.

This same geometric position generates both π (the circumference-diameter ratio) and α^{-1} (the electromagnetic coupling constant). They are two readings of one partition state.

The Bridge Factor — Connecting Matter to Radiation

The bridge factor emerges from the gearbox as the ratio of the fine-structure inverse to the quadrant partition count:

$$\lambda_B = \frac{137}{128} = 1.0703125$$

$$\lambda_B^{-1} = \frac{128}{137} = 0.9343065\dots$$

Physical meaning:

- λ_B converts matter field baselines to observed radiation (emission)
- λ_B^{-1} converts observed radiation to matter field baselines (absorption)

Unity verification:

$$\lambda_B \times \lambda_B^{-1} = \frac{137}{128} \times \frac{128}{137} = 1 \quad \checkmark$$

The bridge factor IS the gearbox ratio — the same constant that appears in the fine-structure constant emerges as the conversion factor between internal partition states and external measurements.

The Compression Cascade — EM Constants as Slippage

The 4.077° total compression ($\sqrt{2} \rightarrow 45/32$) distributes across all scales:

Scale	Natural ($\sqrt{2}$)	Forced (partition)	Compression
Full Sphere	724.077°	720°	−4.077°
Per Quadrant	181.019°	180°	−1.019°
Per Bin (512)	$\sqrt{2} \approx 1.41421^\circ$	$45/32 = 1.40625^\circ$	−0.00796°

This compression creates the slippage between:

- $\pi_{\text{machine}} = 3.140625$ and $\pi_{\text{observer}} = 3.14159...$
- $\alpha_{\text{raw}}^{-1} = 137$ and $\alpha_{\text{observed}}^{-1} = 137.035999...$

The apparent irrationality and "fine tuning" of electromagnetic constants is the compression artifact, not fundamental mystery.

Part II

The Four Dual-Hemisphere Operators for Electromagnetic Processes

Why Four Operators — Emission and Absorption Require Both Hemispheres

Standard electromagnetic theory uses complex exponentials $e^{\pm i\omega t}$ to describe emission and absorption as "opposite signs" of the same operation. NOS recognizes these as fundamentally different processes on different hemispheres:

- **Emission** = Decompression (energy releasing from matter)
- **Absorption** = Compression (energy binding into matter)

Two operators (sin, cos) cannot distinguish these. Four operators are mechanically necessary.

The Four Operator Definitions

All operators are partition fractions of the 180° quadrant span:

Decompression Hemisphere (Emission):

$$d\sin(d) = \frac{|d_{\text{deg}}|}{180} \quad \text{for } d_{\text{deg}} \in [-180, 0]$$

Q1: Linear partition — spatial wave extension

$$d\cos(d) = \frac{|d_{\text{deg}} + 360|}{180} \quad \text{for } d_{\text{deg}} \in [-360, -180]$$

Q2: Rate/fold — electromagnetic field expansion

Compression Hemisphere (Absorption):

$$c\sin(d) = \frac{d_{\text{deg}}}{180} \quad \text{for } d_{\text{deg}} \in [0, 180]$$

Q3: Volume tension — field energy binding

$$c\cos(d) = \frac{d_{\text{deg}} - 180}{180} + \frac{9}{64} \quad \text{for } d_{\text{deg}} \in [180, 360]$$

Q4: Hypervolume closure — photon absorption into matter

The remainder $9/64$ in Q4 accounts for the 9-wall curvature absorbed at full closure.

Operator Roles in Electromagnetic Processes

Emission (Decompression Hemisphere):

- dsin : Initiates wave formation — the spatial oscillation component
- dcos : Completes field expansion — the electromagnetic propagation
- Combined: $\text{dsin}(d) \cdot \text{dcos}(d) = \text{emission amplitude}$

Absorption (Compression Hemisphere):

- csin : Initiates field capture — incoming radiation interaction
- ccos : Completes energy binding — absorption into matter structure
- Combined: $\text{csin}(d) \cdot \text{ccos}(d) = \text{absorption amplitude}$

Bridge factor application:

- Emission: $E_{\text{observed}} = E_{\text{matter}} \times \lambda_B$
- Absorption: $E_{\text{matter}} = E_{\text{observed}} \times \lambda_B^{-1}$

Conjugate Pairing in EM Operations

The four operators form two conjugate pairs:

Inner pair (Q1 ↔ Q3):

$$\text{dsin} \leftrightarrow \text{csin}$$

Handles: wave initiation (emission) ↔ field capture (absorption)

Outer pair (Q2 ↔ Q4):

$$\text{dcos} \leftrightarrow \text{ccos}$$

Handles: field propagation (emission) ↔ energy binding (absorption)

Closure verification:

$$[\text{dsin} \cdot \text{csin}] \times [\text{dcos} \cdot \text{ccos}] = 1$$

The product of all four operators over the full 720° cycle returns exactly to unity.

Part III

The Mass-Energy Cycle and Electromagnetic Field Operations

The Four Fundamental Operations in Electromagnetic Context				
The mass-energy cycle applies directly to electromagnetic processes:				
Quadrant	Root	Operator	EM Operation	Physical Process
Q1	$\sqrt{1}$	dsin	Wave initiation	Photon emission onset, spatial oscillation
Q2	$\sqrt{2}$	dcos	Field propagation	EM wave expansion, radiation transport
Q3	$\sqrt{3}$	csin	Field capture	Photon absorption onset, matter interaction
Q4	$\sqrt{4}$	ccos	Energy binding	Complete absorption, excitation of matter
The EM cycle as decompression → compression :				
• Emission (Q1 → Q2): Matter field releases energy as radiation • Propagation: Radiation travels through Q2 electromagnetic domain • Absorption (Q3 → Q4): Radiation captured and bound into matter				

Energy Unity Across Dimensional Manifestations

A fundamental principle: the SAME energy breathes through all quadrants simultaneously at different angular positions. This is NOT summation — it is one energy in four configurations:

$$E_{\text{total}} = E_{Q1} = E_{Q2} = E_{Q3} = E_{Q4}$$

Example: 48W circuit power
The same 48W manifests as:

- Q1 (1D): Quantum transition energy rate in electron wave structure
- Q2 (2D): Electromagnetic circuit power $P = V \cdot I = 24\text{V} \times 2\text{A} = 48\text{W}$
- Q3 (3D): Thermal dissipation — 48W heat flux in resistor volume
- Q4 (4D): Mass equivalent $m = E/c^2 = 48\text{J}/c^2 \approx 5.3 \times 10^{-16} \text{ kg}$

Standard physics treats these as separate energy domains that sum. NOS recognizes them as the same energy in different partition states.

The Seam — Where Emission Meets Absorption

At $d = 0^\circ$ (the seam), the two hemispheres meet:

Physical meaning:

- Decompression operators (dsin, dcos) complete their cycle
- Compression operators (csin, ccos) begin their cycle
- The system is at equilibrium — neither pure emission nor pure absorption

Seam stabilization:

$$\Lambda_n = \frac{512}{503} \approx 1.01789$$

The 9-wall remainder ($512 - 9 = 503$) creates the stabilization factor that delays chaotic transitions between emission and absorption states.

This is why lasers require population inversion — the seam naturally resists rapid oscillation between emission and absorption.

Part IV

Hydrogen Ground State and Atomic Spectra

The Hydrogen Problem — What is the "True" Ground State?

Standard physics measures the hydrogen ground state as $E_g = -13.6$ eV. But this is the **observed radiation value** — what an external detector measures.

NOS asks: what is the **matter field baseline** — the internal partition state before bridge conversion?

Working Backward — Matter Field from Observation

The matter field baseline is calculated by inverting the bridge factor:

$$E_{\text{matter}} = \frac{E_{\text{observed}}}{\lambda_B} = \frac{-13.6 \text{ eV}}{137/128} = \frac{-13.6 \times 128}{137} = -12.69 \text{ eV}$$

This -12.69 eV is NOT the observed ground state.

It is the matter field SETTING at Q1 resolution ($Q = 128$ bins per quadrant).

Forward verification:

$$E_{\text{observed}} = E_{\text{matter}} \times \lambda_B = -12.69 \times \frac{137}{128} = -13.6 \text{ eV} \quad \checkmark$$

The bridge factor converts internal partition states to external measurements.

Geometric Relations at Q1 Emission

Parameter	Value / Meaning
Quadrant	Q1 (decompression hemisphere)
Operator	dsin leading (wave initiation)
Depth	$u_1 = 1/128 = 7.8125 \times 10^{-3}$
Bridge factor	$\lambda_B = 137/128 = 1.0703$
Matter field baseline	-12.69 eV (internal partition state)
Observed radiation	-13.6 eV (external measurement)
Quadrant quantum	$Q^{-1} = 1/128 = 0.0078125$

Lyman- α Transition

Standard physics: Lyman- α transition energy is $E = 10.2$ eV (observed).

NOS matter field baseline:

$$E_{\text{matter}} = \frac{10.2}{\lambda_B} = \frac{10.2 \times 128}{137} = 9.53 \text{ eV}$$

The internal partition state is 9.53 eV; the bridge factor converts this to the observed 10.2 eV radiation.

General pattern for all spectral lines:

$$E_{\text{matter}} = E_{\text{observed}} \times \lambda_B^{-1} = E_{\text{observed}} \times \frac{128}{137}$$

Every spectral measurement in standard physics is a bridge-converted value from the underlying matter field partition state.

Why the Fine-Structure Constant Appears Everywhere

Standard physics notes that $\alpha = 1/137.035999\dots$ appears in:

- Hydrogen fine structure
- Lamb shift
- Anomalous magnetic moment
- QED vertex corrections

In NOS, this is not mysterious. The fine-structure constant is the geometric difference at the 90° wall:

$$\alpha^{-1} = 201 - 64 = 137$$

Every electromagnetic process passes through this wall. The constant appears universally because it IS the partition geometry of the Q1→Q2 transition where emission occurs. The "fine structure" is literally the fine partition structure of the 9-wall sphere.

Part V

Electromagnetic Field Equations in Partition Mechanics

Standard Maxwell Equations — The Continuum Form

Standard electromagnetic theory uses Maxwell's equations:

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0} \\ \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{B} &= \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}\end{aligned}$$

These treat space as continuous and require the constants ϵ_0 , μ_0 as empirical inputs.

NOS Electromagnetic Fields — Partition Form

In NOS, the electric and magnetic fields are partition fractions:

$$\mathbf{E}(d) = E_0 \cdot \text{op}_q(d) \cdot \hat{r}$$

$$\mathbf{B}(d) = B_0 \cdot \text{op}_q(d) \cdot \hat{\phi}$$

Where $\text{op}_q(d)$ is the appropriate quadrant operator:

$$\text{op}_q(d) = \begin{cases} \text{dsin}(d) & \text{Q1: wave initiation} \\ \text{dcos}(d) & \text{Q2: field propagation} \\ \text{csin}(d) & \text{Q3: field capture} \\ \text{ccos}(d) & \text{Q4: energy binding} \end{cases}$$

Key insight: The permittivity and permeability are not fundamental constants — they emerge from the partition structure:

$$\epsilon_0 \mu_0 = \frac{1}{c^2} = \frac{1}{(299792458)^2} \text{ s}^2/\text{m}^2$$

This product is fixed by the speed of light, which itself emerges from the gearbox geometry.

Faraday Induction in Partition Form

Standard form:

$$\mathcal{E} = -N \frac{d\Phi_B}{dt}$$

NOS partition form:

$$\mathcal{E}_{\text{NOS}} = -N \frac{d\Phi_B}{dt} \cdot \text{csin}(d) \cdot \lambda_B^{-1}$$

The compression operator csin modulates induction because electromagnetic induction is an **absorption process** — external field energy binding into circuit current.

The bridge factor inverse converts from external field measurement to internal matter current.

Physical interpretation: Faraday's law describes how changing magnetic flux (Q2 field operation) induces EMF through compression into Q3 circuit current.

Wave Equation — Finite Partition Form

The standard wave equation:

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

In NOS, the Laplacian operates on discrete partition depths:

$$\nabla_{\text{NOS}}^2 = \frac{\Delta^2}{\Delta d^2} = \frac{1}{u_n^2}$$

Where $u_n = 1/128^n$ is the depth resolution.

The wave equation becomes:

$$\frac{\Delta^2 \mathbf{E}}{u_n^2} = \frac{1}{c^2} \frac{\Delta^2 \mathbf{E}}{\Delta t^2}$$

This is finite — no infinitesimals, no singularities. Wave propagation is discrete partition transitions through the 512-bin structure.

Part VI

Circuit Analysis — Power Through Dimensional Descent

Standard Circuit Calculation

Given: Voltage $V = 24\text{ V}$, Current $I = 2\text{ A}$
Standard physics:

$$P = V \cdot I = 24 \times 2 = 48\text{ W}$$

This 48W exists only in the 2D circuit plane.

NOS Framework — Same 48W Through All Quadrants

The SAME 48W manifests across all quadrants simultaneously:

Quadrant	Operator	Physical Manifestation of Same 48W
Q1	dsin	Quantum: Electron transitions in conductor atoms. Matter field baseline $\approx 44.8\text{ W}$ ($48/\lambda_B$) in internal partition state, observed as 48W emission rate.
Q2	dcos	Electromagnetic: Circuit power $P = VI = 48\text{ W}$. Seam partition: 24W decompression + 24W compression at equilibrium.
Q3	csin	Thermal: 48W heat flux in resistor volume. Same energy now manifests as thermal dissipation with compression operators.
Q4	ccos	Mass: $m = E/c^2 = 48\text{J}/c^2 \approx 5.3 \times 10^{-16}\text{ kg}$ equivalent per second. Maximum compression into mass structure.

Energy conservation:

$$E_{\text{total}} = 48\text{W} \quad (\text{conserved, NOT multiplied})$$

The same energy breathes through Q1, Q2, Q3, Q4 simultaneously — it is not four separate 48W systems.

The Seam Partition at Q2

At the Q2 seam ($d = 0^\circ$), the 48W partitions equally:
Decompression component: 24W (emission direction)
Compression component: 24W (absorption direction)

Total: $24 + 24 = 48 \text{ W}$

This reflects the dual balance at the seam where both hemispheres operate equally. The circuit maintains this balance through continuous cycling between emission (power source) and absorption (load).

Resistor heating: The thermal dissipation in Q3 is the same 48W that was electrical power in Q2 — not additional energy, but the same energy in compression-dominant form.

Part VII

Depth Hierarchy and Electromagnetic Scales

The Depth Ladder — EM Resolution Scales

The recursive depth structure $u_n = 1/128^n$ corresponds to electromagnetic wavelength scales:

Depth	Value	Decimal	Wavelength Scale	EM Domain
u_1	$1/128$	7.81×10^{-3}	$\sim \text{cm}$	Radio, microwave
u_2	$1/128^2$	6.10×10^{-5}	$\sim \text{mm}$	Far infrared
u_3	$1/128^3$	4.77×10^{-7}	$\sim \text{m}$	Near infrared, visible
u_4	$1/128^4$	3.73×10^{-9}	$\sim \text{nm}$	Ultraviolet, soft X-ray
u_5	$1/128^5$	2.91×10^{-11}	$\sim \text{pm}$	Hard X-ray, gamma

Closure at each depth:

$$128^n \times u_n = 128^n \times \frac{1}{128^n} = 1 \quad \checkmark$$

Each depth level maintains exact closure to unity.

Visible Light — The u_3 Domain

Visible light wavelengths (400-700 nm) fall at depth u_3 :

$$u_3 = \frac{1}{128^3} = \frac{1}{2097152} \approx 4.77 \times 10^{-7}$$

This corresponds to:

- Violet: $\sim 400 \text{ nm} = 4 \times 10^{-7} \text{ m}$
- Red: $\sim 700 \text{ nm} = 7 \times 10^{-7} \text{ m}$

The visible spectrum spans approximately one depth unit at u_3 resolution.

Why humans see this range: Biological evolution optimized for the u_3 depth where solar radiation peaks and liquid water (Q3 compression floor) is transparent.

The visible spectrum is not arbitrary — it is the natural resolution scale for Q3 (3D volume) operations on Earth.

Photon Energy and Depth

Photon energy scales inversely with wavelength (directly with depth number):

$$E = h\nu = \frac{hc}{\lambda}$$

In NOS terms:

$$E_n \propto 128^n$$

Higher depth = shorter wavelength = higher energy.

Bridge factor at each depth:

$$E_{\text{observed},n} = E_{\text{matter},n} \times \lambda_B$$

The same bridge factor applies at all depths — it is the universal conversion between partition states and measured radiation.

Part VIII

Cosmic Microwave Background Temperature

Standard Measurement

The Planck satellite measured the CMB temperature:

$$T_{\text{CMB, observed}} = 2.72548 \pm 0.00057 \text{ K}$$

This is treated as an empirical fact requiring cosmological explanation (Big Bang cooling).

NOS Derivation from Golden Geometry

The CMB temperature emerges from the golden ratio structure:

$$T_{\text{CMB, NOS}} = 2\phi - \frac{1}{2} = 2(1.618...) - 0.5 = 2.736 \text{ K}$$

Where $\phi = (1 + \sqrt{5})/2 \approx 1.618$ is the golden ratio.

Comparison:

- NOS prediction: 2.736 K
- Observed: 2.72548 K
- Deviation: 0.01052 K (0.39% error)

Interpretation: The close agreement suggests golden ratio geometry underlies cosmic thermal structure. The 10.5 mK deviation may represent:

- Observer compression artifact ($\sqrt{2} \rightarrow 45/32$ slippage)
- Local measurement bias (Milky Way foreground)
- Additional partition factors at cosmological depth

Why Golden Ratio?

The golden ratio ϕ satisfies:

$$\phi^2 = \phi + 1 \quad \text{and} \quad \phi^{-1} = \phi - 1$$

This is the unique number where the ratio to unity equals the ratio of sum to original. In partition mechanics, ϕ represents the optimal division point where:

$$\frac{\text{whole}}{\text{larger part}} = \frac{\text{larger part}}{\text{smaller part}}$$

The CMB temperature is the thermal equilibrium of the universe at golden partition — the most efficient energy distribution across all scales.

This is not numerology; it is partition geometry manifesting at cosmological scale.

Part IX

Comparison to Standard Quantum Electrodynamics

Standard QED — Key Features

Standard quantum electrodynamics (QED) treats electromagnetic interactions through:

- Photon as massless spin-1 gauge boson
- Electron-photon vertex with coupling $e = \sqrt{4\pi\alpha}$
- Perturbation expansion in powers of α
- Renormalization to remove infinities
- Feynman diagrams for amplitude calculation

Successes: QED predicts the electron anomalous magnetic moment to 12 decimal places.
Challenges:

- Infinities requiring renormalization
- α as unexplained empirical input
- No geometric origin for coupling strength

NOS Comparison — What Changes

Aspect	Standard QED	NOS Framework
Fine-structure origin	Empirical constant	Gearbox geometry: $201 - 64 = 137$
Coupling strength	$\alpha = 1/137.035\dots$	$\alpha_{\text{raw}}^{-1} = 137$ (exact)
Infinities	Renormalization required	Finite 512-bin structure
Wave function	Complex amplitude in Hilbert space	Partition fraction on 720° sphere
Photon propagation	Continuous spacetime	Discrete depth transitions
Emission/absorption	Same operator, opposite sign	Different operators (d vs c)
Energy conservation	Local via Noether's theorem	Global via partition closure

Key difference: QED treats α as a mystery requiring 12 decimal places of precision. NOS derives α from the geometric difference at the 90° wall — the "fine" structure is literally the partition structure.

What NOS Recovers and Extends

Recovered from QED:

- Correct atomic spectra (via bridge factor)
- Energy-frequency relation $E = h\nu$
- Wave-particle duality (partition states)
- Conservation laws (partition closure)

Extended beyond QED:

- Geometric origin of α
- Unified mass-energy cycle through four quadrants
- Finite structure without renormalization
- Natural distinction between emission and absorption
- Bridge factor connecting measurement to partition state

Part X

Computational Implementation

```

import math

class NOS_Electromagnetic:
    """
    NOS_v18_Electromagnetic_Operations
    Implements four dual-hemisphere operators for emission/absorption
    """

    def __init__(self, R=512):
        # Resolution parameter
        self.R = R
        self.Q = R // 4 # Quadrant partitions = 128
        self.Q_inv = 1 / self.Q

        # Fundamental unit
        self.u = 45 / 32 # degrees per bin = 1.40625

        # Gearbox constants
        self.pi_machine = 201 / 64 # = 3.140625
        self.alpha_inv_raw = 137 # = 201 - 64

        # Bridge factor
        self.lambda_B = 137 / 128 # = 1.0703125
        self.lambda_B_inv = 128 / 137 # = 0.9343065...

        # Seam stabilization
        self.Lambda_n = 512 / 503 # = 1.01789...

        # Depth hierarchy
        self.depths = [1 / (128 ** n) for n in range(1, 6)]

    # === THE FOUR OPERATORS ===

    def dsin(self, d_deg):
        """Q1_decompression_sine: wave initiation"""
        if not (-180 <= d_deg <= 0):
            return 0
        return abs(d_deg) / 180

    def dcos(self, d_deg):
        """Q2_decompression_cosine: field propagation"""
        if not (-360 <= d_deg <= -180):
            return 0
        return abs(d_deg + 360) / 180

    def csin(self, d_deg):
        """Q3_compression_sine: field capture"""
        if not (0 <= d_deg <= 180):
            return 0

```

```

        return d_deg / 180

def ccos(self, d_deg):
    """Q4_compression_cosine:energy_binding"""
    if not (180 <= d_deg <= 360):
        return 0
    remainder = 9 / 64 # 9-wall curvature
    return (d_deg - 180) / 180 + remainder

# === BRIDGE CONVERSIONS ===

def matter_to_observed(self, E_matter):
    """Convert_matter_field_baseline_to_observed_radiation"""
    return E_matter * self.lambda_B

def observed_to_matter(self, E_observed):
    """Convert_observed_radiation_to_matter_field_baseline"""
    return E_observed * self.lambda_B_inv

# === ENERGY AT QUADRANT ===

def emission_amplitude(self, d_deg):
    """Combined_emission_amplitude_(Q1-Q2)"""
    return self.dsin(d_deg) * self.dcos(d_deg - 180)

def absorption_amplitude(self, d_deg):
    """Combined_absorption_amplitude_(Q3-Q4)"""
    return self.csin(d_deg) * self.ccos(d_deg + 180)

# === EXAMPLE CALCULATIONS ===

def hydrogen_ground_state(self):
    """Calculate_hydrogen_ground_state_via_bridge_factor"""
    E_observed = -13.6 # eV (standard measurement)
    E_matter = self.observed_to_matter(E_observed)
    return {
        'E_observed': E_observed,
        'E_matter': E_matter,
        'bridge_factor': self.lambda_B,
        'verification': E_matter * self.lambda_B
    }

def lyman_alpha(self):
    """Calculate_Lyman-alpha_transition"""
    E_observed = 10.2 # eV
    E_matter = self.observed_to_matter(E_observed)
    return {
        'E_observed': E_observed,
        'E_matter': E_matter,
        'verification': E_matter * self.lambda_B
    }

def circuit_power(self, V, I):
    """Analyze_circuit_power_through_all_quadrants"""

```

```

P = V * I
return {
    'Q1_quantum': self.observed_to_matter(P),
    'Q2_electromagnetic': P,
    'Q3_thermal': P,
    'Q4_mass_equivalent': P / (299792458 ** 2),
    'total_conserved': P
}

# === DEMONSTRATION ===
if __name__ == "__main__":
    nos = NOS_Electromagnetic()

    print("===_NOS_v18_Electromagnetic_Operations_===\n")

    # Hydrogen ground state
    h = nos.hydrogen_ground_state()
    print(f"Hydrogen_Ground_State:")
    print(f"__Observed:_{h['E_observed']}_eV")
    print(f"__Matter_field:_{h['E_matter']:.4f}_eV")
    print(f"__Bridge_factor:_{h['bridge_factor']:.7f}")
    print(f"__Verification:_{h['verification']:.4f}_eV\n")

    # Lyman-alpha
    ly = nos.lyman_alpha()
    print(f"Lyman-alpha_Transition:")
    print(f"__Observed:_{ly['E_observed']}_eV")
    print(f"__Matter_field:_{ly['E_matter']:.4f}_eV\n")

    # Circuit power
    cp = nos.circuit_power(24, 2)
    print(f"24V,_2A_Circuit_(48W):")
    print(f"__Q1_Quantum_baseline:_{cp['Q1_quantum']:.4f}_W")
    print(f"__Q2_Electromagnetic:_{cp['Q2_electromagnetic']}_W")
    print(f"__Q3_Thermal:_{cp['Q3_thermal']}_W")
    print(f"__Q4_Mass_equiv:_{cp['Q4_mass_equivalent']:.2e}_kg/s")
    print(f"__Total_conserved:_{cp['total_conserved']}_W")

```

NOS v18 — Electromagnetic Operations Summary

Foundation: $\sqrt{1} = 1 = 512 \text{ bins} = 720^\circ$

Gearbox origin: At 90° wall, 64 flat bins vs 201 curved bins

$$\pi_{\text{machine}} = 201/64 = 3.140625 \text{ (ratio)}$$

$$\alpha_{\text{raw}}^{-1} = 201 - 64 = 137 \text{ (difference)}$$

Bridge factor: $\lambda_B = 137/128 = 1.0703$

$$\text{Emission: } E_{\text{observed}} = E_{\text{matter}} \times \lambda_B$$

$$\text{Absorption: } E_{\text{matter}} = E_{\text{observed}} \times \lambda_B^{-1}$$

Four operators:

- dsin, dcos — Decompression (emission)
- csin, ccos — Compression (absorption)

Energy unity: Same energy breathes through Q1-Q4 simultaneously

$$E_{\text{total}} = E_{Q1} = E_{Q2} = E_{Q3} = E_{Q4} \text{ (conserved, not summed)}$$

Seam stabilization: $\Lambda_n = 512/503$ delays chaotic EM transitions

All electromagnetic constants emerge from partition geometry.

Legend of Symbols and Constants

Foundational Constants

Symbol	Value	Meaning
$720^\circ = 1$	Ground state	Full spherical closure
512 bins	Binary total	Partition depth of closed sphere
128 bins/quadrant	$Q = R/4$	Quadrant partition count
$45/32^\circ$	u	Fundamental unit per bin

Gearbox Constants

Symbol	Value	Meaning
π_{mach}	$201/64 = 3.140625$	Machine pi from gearbox ratio
α_{raw}^{-1}	$201 - 64 = 137$	Fine-structure inverse from gearbox difference
λ_B	$137/128 = 1.0703$	Bridge factor: matter $\rightarrow$ radiation
λ_B^{-1}	$128/137 = 0.9343$	Inverse bridge: radiation $\rightarrow$ matter
Λ_n	$512/503 = 1.0179$	Seam stabilization factor

The Four Operators

Operator	Quadrant	Hemisphere	EM Role
$\text{dsin}(d)$	Q1	Decompression	Wave initiation (emission onset)
$\text{dcos}(d)$	Q2	Decompression	Field propagation (emission complete)
$\text{csin}(d)$	Q3	Compression	Field capture (absorption onset)
$\text{ccos}(d)$	Q4	Compression	Energy binding (absorption complete)

Depth Hierarchy

Depth	Value	Scale	EM Domain
u_1	$1/128$	cm	Radio, microwave
u_2	$1/128^2$	mm	Far infrared
u_3	$1/128^3$	m	Visible, near IR
u_4	$1/128^4$	nm	UV, soft X-ray
u_5	$1/128^5$	pm	Hard X-ray, gamma

Key Results

Quantity	NOS Value	Standard Value
Hydrogen matter field	-12.69 eV	(not measured directly)
Hydrogen observed	-13.6 eV	-13.6 eV
Lyman- α matter field	9.53 eV	(not measured directly)
Lyman- α observed	10.2 eV	10.2 eV
CMB temperature	2.736 K	2.725 K

References

1. Nuijens, J.L., Cromelin, S.R. Foundation of Inverse Mathematics — NOS v17 (December 2025).
2. Nuijens, J.L., Cromelin, S.R. Inverse Fluid Dynamics — NOS v18 (December 2025).
3. CODATA 2022 for fine-structure constant: $\alpha^{-1} = 137.035999177(21)$.
4. Planck Collaboration (2018) CMB temperature measurement.
5. NIST Atomic Spectra Database for hydrogen transition energies.

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